

MINI PROJECT REPORT

Number Guessing Game using Tkinter

1. Introduction

This mini project focuses on developing a Number Guessing Game using Python's Tkinter library. Tkinter is a standard GUI (Graphical User Interface) toolkit in Python that allows developers to create desktop applications easily. The aim of this project is to create an interactive game where the user guesses a randomly generated number.

2. Objectives

- To design a GUI-based game using Tkinter
- To generate random numbers and implement guessing logic
- To understand event handling in Python
- To improve user interaction with graphical elements

3. Tools & Technologies Used

- Programming Language: Python
- Library: Tkinter
- IDE: IDLE / VS Code / PyCharm

4. System Requirements

Hardware Requirements:

- Computer with minimum 4GB RAM

Software Requirements:

- Python 3.x
- Tkinter (comes pre-installed with Python)

5. Project Description

The Number Guessing Game is a GUI-based application where the system generates a random number between 1 and 100. The user is asked to guess the number through an input field. The application provides hints such as "Too High" or "Too Low" based on the user's input. The game continues until the correct number is guessed.

6. Modules Used

- tkinter: To create GUI components
- random: To generate random numbers
- messagebox: To display alerts and messages

7. Algorithm

1. Start the program
2. Import required modules
3. Generate a random number
4. Create the main window
5. Add input field and buttons
6. Accept user guess
7. Compare guess with random number
8. Display result (Too High / Too Low / Correct)
9. Repeat until correct guess
10. End

8. Program Code

```
import tkinter as tk
from tkinter import messagebox
import random

number = random.randint(1, 100)

def check_guess():
    try:
        guess = int(entry.get())
        if guess < number:
            result_label.config(text="Too Low!")
        elif guess > number:
            result_label.config(text="Too High!")
        else:
            messagebox.showinfo("Success", "Correct Guess!")
    except:
        messagebox.showerror("Error", "Enter a valid number")

root = tk.Tk()
root.title("Number Guessing Game")
root.geometry("300x200")

tk.Label(root, text="Guess a number (1-100)").pack(pady=10)

entry = tk.Entry(root)
entry.pack()

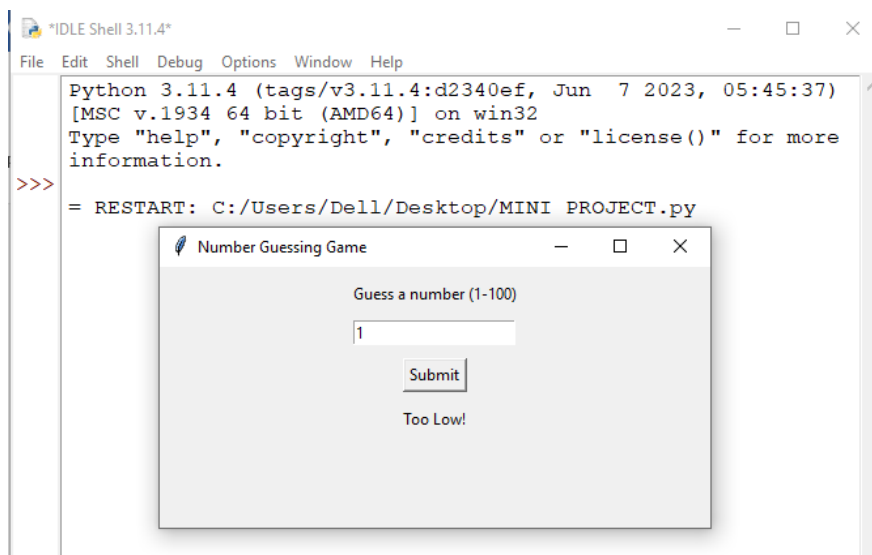
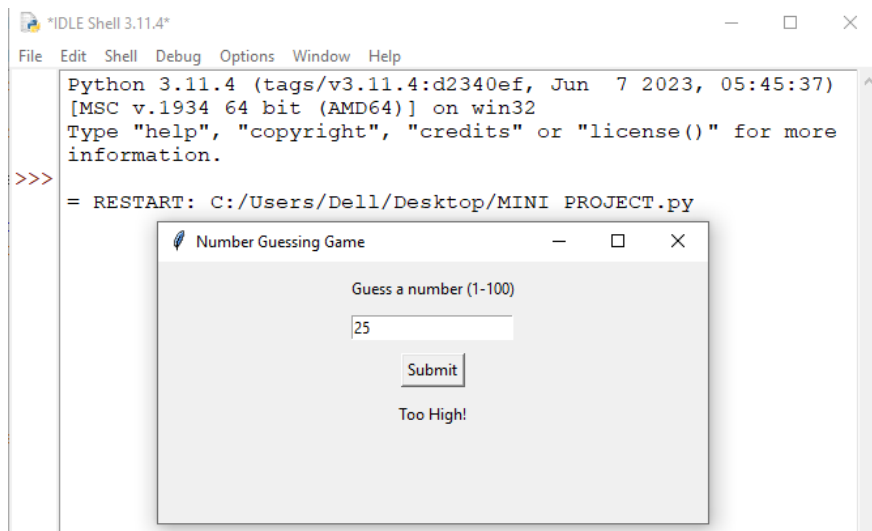
tk.Button(root, text="Submit", command=check_guess).pack(pady=10)
```

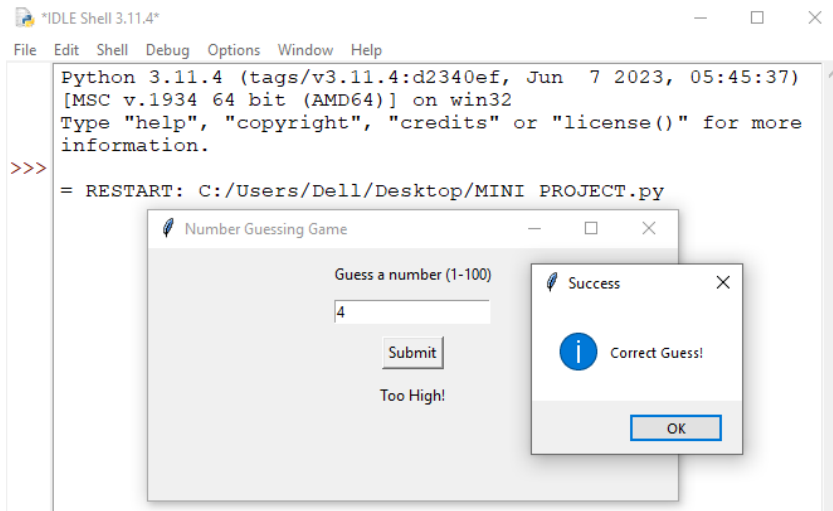
```
result_label = tk.Label(root, text="")  
result_label.pack()
```

```
root.mainloop()
```

9. Output

- A window appears asking the user to guess a number
- User enters a number and clicks submit
- The system displays whether the guess is too high, too low, or correct





10. Advantages

- Simple and interactive game
- Helps understand GUI and logic building
- Beginner-friendly project

11. Limitations

- Limited features
- No score tracking
- No difficulty levels

12. Future Enhancements

- Add score tracking system
- Introduce difficulty levels
- Add timer feature

13. Conclusion

This project demonstrates how to build a simple interactive game using Tkinter. It helps beginners understand GUI development, event handling, and logical thinking in Python.

14. References

- Python Documentation
- Tkinter Tutorials